

No Single Winner in the Electric Vehicle Battery Race



The electric vehicle battery technology landscape has undergone major transformation across decades, largely driven by the inherent limitations of earlier chemistries. Each generation of batteries has not only improved performance but also revealed new constraints, acting as a catalyst for the next wave of innovation focused on:

- Energy density
- Cost
- Safety
- Lifecycle efficiency

Lead-Acid Battery Era

Rewind to the early 1990s, when lead-acid batteries were among the first technologies used in battery electric vehicles (BEVs). However, their adoption declined owing to lower energy density, which restricted driving range, and a relatively short cycle life. This limited durability stemmed from degradation of the internal plates over repeated charging and discharging within the battery's acidic electrolyte.

Further, lead-acid batteries typically require long charging times (often around 8 to 16 hours), which makes them less practical for modern EV applications. Since lead-acid batteries are an old technology, there isn't much room left to improve them further because they've already reached most of their performance limits.

Another battery technology - nickel metal hydride (NiMH) - emerged in the automotive sector in the late nineties and gained strong traction in hybrid electric vehicles, with Toyota Motor Corporation deploying it across models such as the Toyota Prius and Toyota Camry Hybrid. NiMH batteries offer higher energy density, improved cycle life, and strong recyclability, making them well-suited for the frequent charge–discharge cycling required in hybrid systems. They are also relatively less toxic and can be recycled effectively.

However, NiMH batteries did not quite gain market acceptance because they tend to generate more heat under high-rate charging and heavy load conditions, and their efficiency declines under such stress. These thermal and performance constraints made them less suitable for full battery electric vehicles, where range, fast charging, and sustained efficiency are critical.

Proliferation of Lithium-Based Chemistries

These constraints accelerated the shift toward lithium-based chemistries, which emerged as the dominant force in automotive electrification in the late 1990s and continue to anchor the EV industry today. However, the lithium-ion technology is no longer a single unified pathway - instead, it has evolved into a diversified ecosystem of chemistries optimized for distinct use cases.

Lithium iron phosphate (LFP) batteries have transitioned from a cost-driven alternative to a mainstream solution across electric vehicles, including mass-market passenger EVs, entry-level vehicles, two-wheelers, and commercial fleet applications. Their appeal now extends beyond affordability, to structural safety, long cycle life, and steadily improving energy density, supported by innovations such as cell-to-pack and blade-style architectures. In contrast, traditional lithium nickel manganese cobalt oxide (NMC) batteries continue to dominate long-range and premium electric vehicles, where high energy density is non-negotiable.

It is important to observe that even within these chemistries, the continuous reduction of cobalt content and higher nickel concentrations highlight a broader industry shift toward lower cost structures and more ethical supply chain alignment, rather than purely pursuing energy density improvements.

New Variants Come into Play

Further, the notion of lithium-ion as a single, linear technology pathway has become increasingly outdated. New variants such as lithium manganese iron phosphate (LMFP) are emerging to bridge the performance gap between LFP and NMC, while retaining cost efficiency. In parallel, advancements in silicon-based anodes and high-voltage electrolyte systems are pushing the

operational boundaries of existing lithium-ion systems, delaying the need for entirely new battery technologies.

At the same time, sodium-ion batteries are transitioning from laboratory development to early commercialization, especially in low-speed electric vehicles, two-wheelers, and stationary energy storage systems. The strategic value of these batteries lies in reducing exposure to lithium supply concentration and price volatility. However, their lower energy density is likely to limit sodium-ion batteries to specific application segments rather than broad-based adoption in the near term.

Solid-state batteries continue to attract significant investment and attention, but they remain largely in the prototype and pilot stages. They face key challenges including manufacturing complexity, interface stability, and high production costs, despite their widely recognized potential advantages in safety and energy density.

Similarly, lithium titanate (LTO) batteries remain relevant only in niche applications that require extremely ultra-fast charging and very long cycle life, such as high-utilization fleets and specialized industrial use cases, but they are unlikely to scale into mainstream passenger EVs.

Conclusion

Across all battery chemistries, the focus is shifting away from pure performance gains toward system-level optimization, including thermal management, pack architecture, charging integration, and end-of-life recycling economics. At the same time, resource geopolitics is reshaping supply chains, with expanding lithium production and recycling ecosystems easing earlier constraints, while cobalt dependency continues to decline structurally.

This is reinforcing a broader structural shift in the EV industry - there will be no single winning battery chemistry. Instead, the market is evolving into a multi-technology system architecture, where each chemistry survives based on fit-for-purpose economics and operational requirements rather than absolute superiority. Competitive advantage is increasingly defined by optimal battery-system design for each mobility application, rather than dominance of a single chemistry.